Expert, crowd and machine: comparing LLM and human classification distributions of strategic thinking

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Motivation

Beyond a single label

For text classification with LLMs, we currently ask for *one label*, but we can also get a close approximation of the model's *probability distribution* over the classification categories.

We can get this **additional information** for **no additional cost**.

It provides **the model's degree of uncertainty** over the task (data + model uncertainty).

→ Can we leverage this additional information to "improve" text classification using LLMs?

details

Project in a nutshell

We have:

- Text messages generated in a lab experiment by subjects (Beauty Contest)
- Benchmark: Classification of this text by two PhD students (experts).
- Classification of this text by a group of non-expert annotators (crowd).

We compare:

- LLM classification with these two groups of annotators.

Terminology:

- Model's distribution over classification categories $\rightarrow$ uncertainty of the model
- Category with highest assigned probability $\rightarrow$ confidence of the model's classification

Research Questions

RQ1: Do LLMs mirror human uncertainty?

(a) How closely do LLM's uncertainty aligns with the experts' uncertainty?
(b) How closely do LLM's uncertainty aligns with crowd's uncertainty?

RQ2: Does uncertainty track accuracy?

(a) Is the model's confidence calibrated to its accuracy?
(b) Can LLM entropy be leveraged for hybrid classification (allocating high-uncertainty items to the crowd) in a way that delivers meaningful accuracy gains?

RQ3: Reproducibility

(a) How stable are LLM predictions across runs?
(b) Do model uncertainty helps us identify low-reproduciblity items?

Data - Burchardi and Penczynski (2014)

Beauty contest game, $a \in [0, 100]$, $p = 2/3$

Text data generated by intra-team communication. ▸ more

78 messages classified by 2 PhD students (expert annotators)

Level-0 belief mean (belief)

- 9 categories: none, 0-15, 16-25, 26-35, 36-45, 46-55, 56-65, 66-75, 76-100
- Instructed to provide a single category classification.

Level of reasoning (level)

- 8 categories: NA, 0, 1, 2, 3, 4, 5, eq
- Instructed to provide a lower and an upper bound.

Procedures

AMT Annotation

Classification done by Mechanical Turkers.

Categories: same categories used for RAs are used.

42-43 classifications per message. more

Almost the same instructions provided to RAs are provided (with an initial layman explanation for what strategic thinking is).

Asked to provide a single category classification for both beliefs and levels.

Procedures

LLM Classification

GPT-3.5, GPT-4

GPT-4o-mini, GPT-4o

5 runs with each model

Almost the same instructions provided to RAs are provided as the prompt.

Template

Accuracy

	3.5	4	4o-mini	4o	MT
MT	.44 ±.00	**.70** ±.01	.55 ±.01	.68 ±.03	1
RA	.32 ±.00	**.69** ±.01	.46 ±.01	.63 ±.01	.64

Table: Belief Accuracy (Mean ± SD)

	3.5	4	4o-mini	4o	MT
MT	.66 ±.01	**.76** ±.02	.72 ±.01	.68 ±.03	1
RA	.69 ±.00	**.78** ±.01	.76 ±.01	.71 ±.01	**.78**

Table: Level Accuracy (Mean ± SD)

graphs example

Entropy

	3.5	**4**	**4o-mini**	**4o**	**MT**	**RA**
Belief	.24 $\pm$.01	.12 $\pm$.02	.18 $\pm$.03	.09 $\pm$.01	.28	–
Level	.18 $\pm$.01	.10 $\pm$.02	.09 $\pm$.02	.11 $\pm$.02	.35	.06

Table: Normalized Entropy (Mean $\pm$ SD)

- LLMs are more certain than the crowd.
- LLMs are more uncertain than the experts.
- Larger models exhibit lower degree of uncertainty.

JSD

SRC

Entropy Correlation

	3.5	4	4o-mini	4o
MT	.38 ±.01	.64 ±.03	.36 ±.03	.71 ±.00

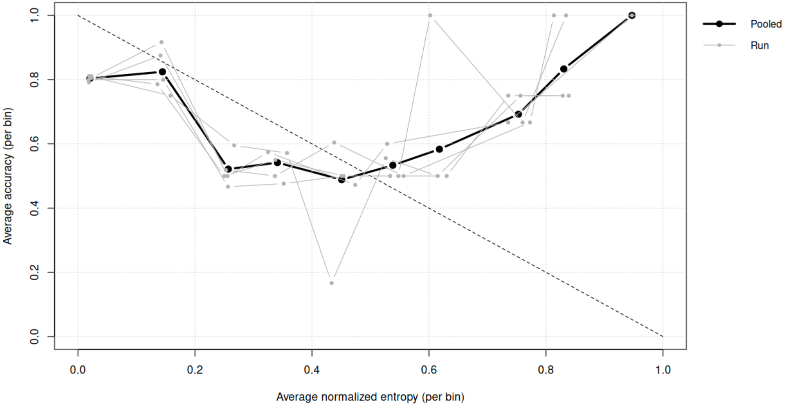
Table: Belief Entropy Correlation (Mean ± SD)

	3.5	4	4o-mini	4o	MT
MT	.43 ±.01	.38 ±.01	.35 ±.03	.34 ±.02	1
RA	.17 ±.00	.27 ±.01	.24 ±.03	.20 ±.01	.22

Table: Level Entropy Correlation (Mean ± SD)

Accuracy ∼ Entropy

GPT 4: Level Accuracy ∼ Total Entropy

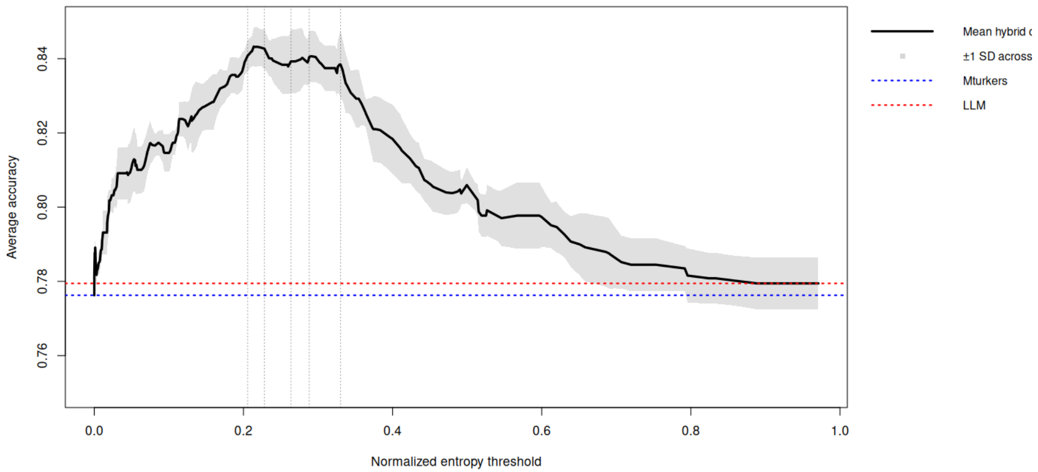


AMT

Confidence

Hybrid Classification

GPT 4: Level Accuracy and Total Entropy



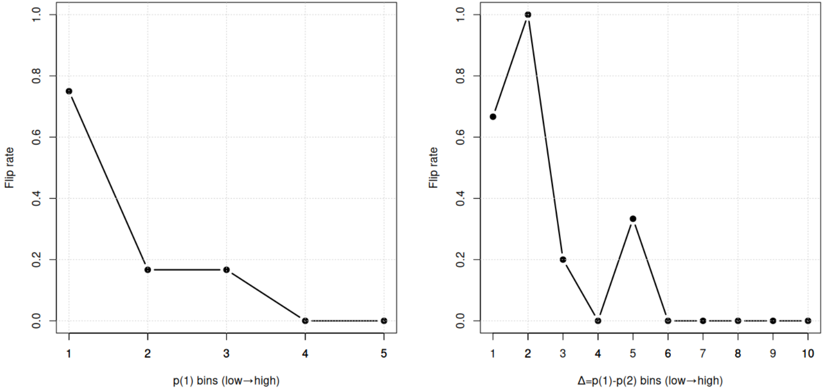
Hybrid Classification

GPT 4: Level Accuracy and Total Entropy

- Optimal Entropy treshold across runs: 0.2-0.3
- Percent classified by GPT 4 across runs: 60%-70%
- Percentage gain vs only LLM classification: 5-7 ppt
- Percentage gain vs only AMT classification: $\sim$ 7 ppt

Reproducibility ∼ Confidence

GPT 4: Level classification



TAR

Recap

- Model's uncertainty moderately correlates with human uncertainty.
- Model's uncertainty moderately correlates with its level of accuracy and this can be leverage in a meaningful way to improve upon both human and LLM accuracy via hybrid classification.
- Model's uncertainty is indicative of its output's stability, and it may be used as a way to signal whether the model's output is reproducible.

Thank you!

Questions?

How LLMs Work Under the Hood (in one slide)

- Model computes **logits** $z_i \in (-\infty, \infty)$ for each token i in its vocabulary.
- Apply **softmax with temperature** $T \in (0, \infty)$:

$$P(i, T) = \frac{e^{z_i / T}}{\sum_{j \in V} e^{z_j / T}}$$

- One token is sampled from the resulting distribution.

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Top-k LogProbs

Some LLM APIs can return the K highest values of the logarithm of predicted token probabilities for the model's next-token prediction (top-k logProbs).

GPT API generates top-k logProbs by:

1. Applying the softmax function with $T = 1$ to logits (z_i):

$$L_{\text{api}}(i) = \log\left(\frac{e^{z_i}}{\sum_{j \in V} e^{z_j}}\right)$$

2. Returning only the k highest L_{api} values (top-k logProbs).

Using top-k logProbs, one can approximate the true probabilities. more

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Top LogProbs + Constrained Output Space

For N classification categories :

1. Limit the token space to N classification categories using JSON Schema output format. more
2. Request top-k logProbs (ideally $K \geq N$).
3. Approximate classification distribution. more

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Limitation

There is a limit to the number top-k logProbs: K_{max}

For GPT 3.5 and 4: $K_{max} = 5$

For GPT 4o-mini and 4o: $K_{max} = 20$

If $N > K_{max}$,

Set an upper bound probability on categories not provided in top-k logProbs. more

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$P(i, T) \approx P_{\text{api}}(i, T)$

- True probabilities (V is the entire set of LLM's vocabulary):

$$P(i, T) = \frac{e^{z_i / T}}{\sum_{j \in V} e^{z_j / T}}$$

- Top-k logProbs (K is the set of tokens with the highest k log probabilities):

$$L_{\text{api}}(i) = \log \left(\frac{e^{z_i}}{\sum_{j \in V} e^{z_j}} \right)$$

- Using top-k logProbs, one can generate probabilities for any temperature T as:

$$P_{\text{api}}(i, T) = \frac{e^{L_{\text{api}}(i) / T}}{\sum_{k \in K} e^{L_{\text{api}}(k) / T}}$$

$P(i, T) \approx P_{\text{api}}(i, T)$

Let $\epsilon = \sum_{j \in V \setminus K} e^{L_{\text{api}}(j)/T}$

Then, with a bit of algebra, one can show that:

$$P(i, T) = \frac{e^{L_{\text{api}}(i)/T}}{\sum_{k \in K} e^{L_{\text{api}}(k)/T} + \epsilon}$$

Hence:

$$\lim_{\epsilon \to 0} P(i, T) = P_{\text{api}}(i, T)$$

In words, if probability mass on other tokens (not top-k) is small enough, then one can reliably construct the true probability distribution for any given T using the top-k logProbs provided by the API.

Constraining Output Token Space

One can constrained the output space of the of the LLM via:

1. **Prompt**
   - Instruct the model to only output token from a specific set.
   - Effective as long as the model is good at instruction following
   - Even GPT-3.5 can follow such instructions (lower bound)
2. **Structured Output - Schema**
   - Define the allowed format (JSON) and set of vocabulary the model can output (token representation of the categories).
   - The model then sets disallowed tokens logits to $-\infty$.
   - API feature available with most providers.

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How to approximate?

1. Convert logProbs to probabilities (e^{l_k}).

2. Assign a minimum value for the missing $N - K$ categories (p_j):

   - $P_{\text{assigned}} = \sum_{k \in K} p_k$
   - $P_{\text{missing}} = 1 - P_{\text{assigned}}$
   - $P_{\text{min}} = \min_{k \in K} p_k$
   - $p_j = \min(P_{\text{missing}}, P_{\text{min}})$

3. Normalize over the entire set.

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Intra-team Communication

- Team of 2 players that play as one entity, aligned incentives
- Simultaneous exchange of individual proposals and messages
- Team action: Draw from the 2 final decisions

→ Incentivized written account of individual reasoning

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Data Collection

42-43 classifications per message, 3350 in total.

59 workers, 23 with whole sets of 78 classifications.

Data collected in July 1 and 5, 2014

Gathered in 5 days open work window

27 seconds median time spent on task, 140 characters median message length

$0.15 per HIT, $8.94 per hour, payment upon our approval

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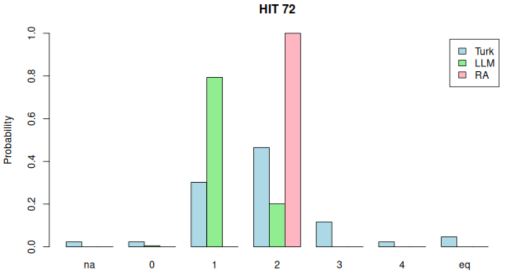
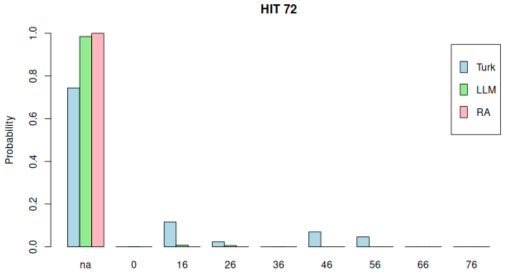
Example

GPT-4



(a) Level classification.

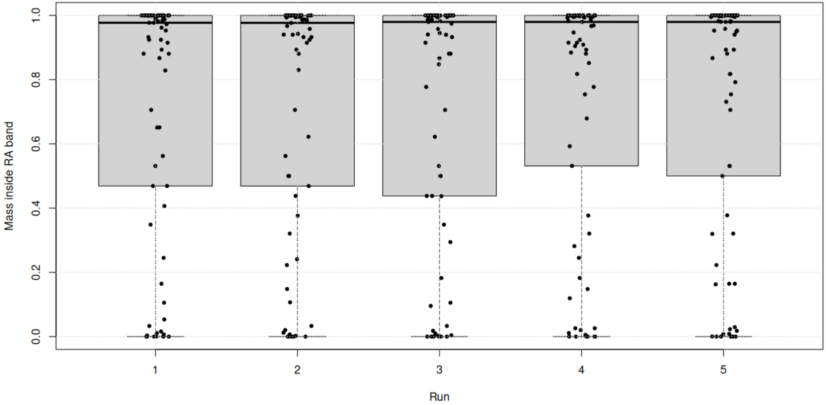


(b) Level-0 belief classification.

Figure: "I assume that most teams will choose low initial numbers, as they will be aiming to undercut the other teams, being closer to the 2/3 target point By choosing 20, i think that it will be high to be above the initial low numbers, whilst not being so astronomically high as to rule us out entirely."

Accuracy

GPT 4



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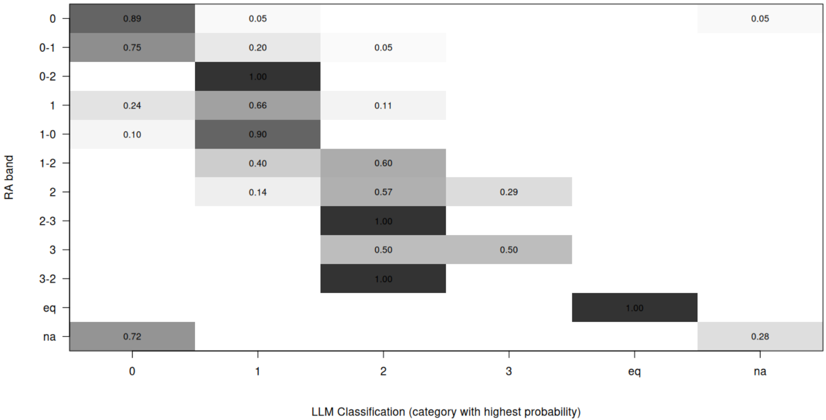
Accuracy

GPT 4



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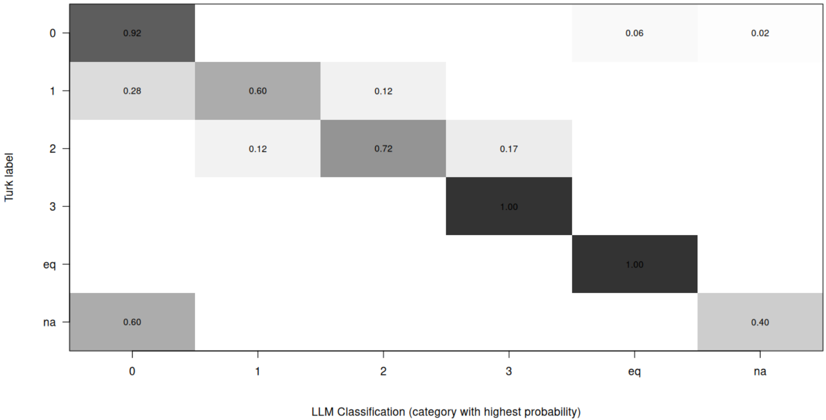
Accuracy

GPT 4



Jensen-Shannon Divergence

	3.5	4	4o-mini	4o
MT	.42 ±.01	.19 ±.02	.35 ±.03	.20 ±.01

Table: Belief JSD (Mean ± SD)

	3.5	4	4o-mini	4o	MT
MT	.17 ±.00	.18 ±.01	.18 ±.01	.20 ±.01	0
RA	.29 ±.01	.22 ±.02	.27 ±.02	.29 ±.02	.21

Table: Level JSD (Mean ± SD)

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Spearman Rank Correlation

	3.5	**4**	**4o-mini**	**4o**
MT	.48 ±.01	.72 ±.03	.56 ±.03	.63 ±.02

Table: Belief SRC (Mean ± SD)

	3.5	**4**	**4o-mini**	**4o**	**MT**
MT	.80 ±.01	.83 ±.02	.78 ±.01	.81 ±.01	1
RA	.55 ±.00	.59 ±.01	.53 ±.01	.53 ±.01	.65

Table: Level SRC (Mean ± SD)

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Entropy ~ Accuracy

	3.5	4	4o-mini	4o
False	.32	.22	.25	.20
True	.14	.07	.13	.04
Signif.	***	***	***	***

Table: Belief Entropy by Accuracy

	3.5	4	4o-mini	4o
False	.23	.12	.10	.10
True	.16	.10	.09	.12
Signif.	.	.	.	.

Table: Level Entropy by Accuracy

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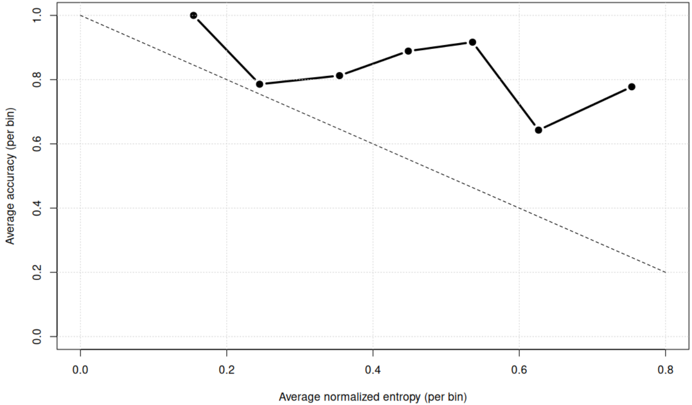
Accuracy ∼ Entropy

AMT: Level Accuracy ∼ Total Entropy



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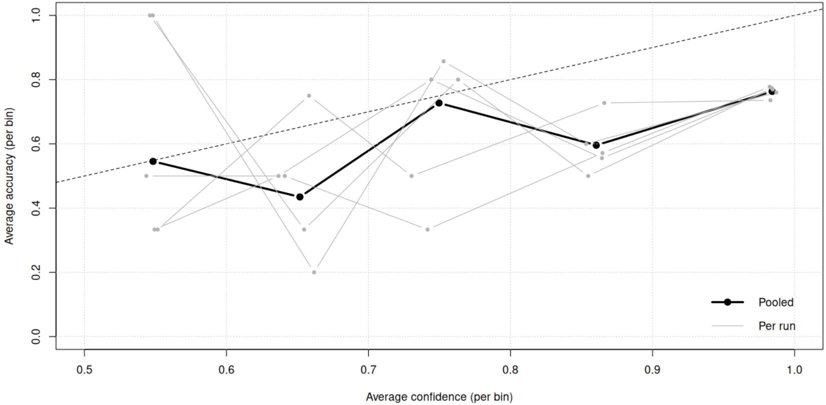
Accuracy ∼ Confidence

GPT 4: Level Accuracy ∼ Confidence



AMT

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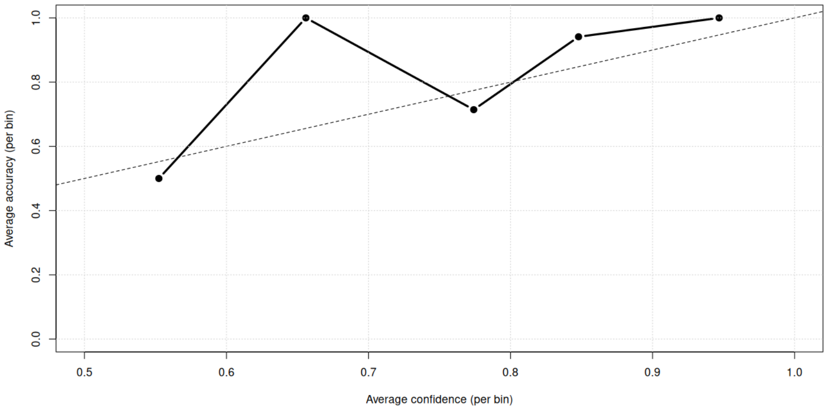
Accuracy ∼ Confidence

AMT: Level Accuracy ∼ Confidence



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Total Agreement Rate

	3.5	**4**	**4o-mini**	**4o**
Belief	.992 (3)	.982 (6)	.938 (15)	.985 (4)
Level	.985 (3)	.982 (5)	.979 (5)	.982 (5)

Table: TAR (number of flips)

Given a sample of 5, TAR $\in \{0.6,\ 0.8,\ 1.0\}$

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TAR $\sim$ Entropy

	3.5	4	4o-mini	4o
Belief	-.23	-.49	-.63	-.48
Level	-.43	-.52	-.42	-.47

Table: Correlation between TAR and Entropy

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TAR $\sim$ Top 2 Prob. Diff.

	3.5	**4**	**4o-mini**	**4o**
Belief	.32	.59	.71	.59
Level	.41	.65	.54	.56

Table: Correlation between TAR and Top2Diff

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Top 2 Prob. Diff.

	3.5	4	4o-mini	4o
Belief	.67	.83	.71	.86
Level	.73	.84	.84	.80

Table: All Cases (Mean)

	3.5	4	4o-mini	4o
Belief	.15 $\pm .03$	.28 $\pm .07$	.21 $\pm .11$	.20 $\pm .11$
Level	.09 $\pm .05$	.17 $\pm .07$	.32 $\pm .20$	0.20 $\pm .13$

Table: Flipped Cases (Mean $\pm$ SD)

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Prompt Template

General Task
- Classify <X> in <E>

Identity
- Act as a behavioral economist specialized in text classification, concept <C> and game <G>

Context
- <Game mechanics>
- <Experimental design/Decision Process>
- <Theory>
- ...

Classification Task
- Classify <X> as <Y> given criteria: <Y_1, Y_2, ...>
- ...

Figure: General Prompt Structure - Part 1

Prompt Template

```
# Classification Coding
- Code <X> as <Z> if it is classified as <Y>
- ...

# Constraint(s)
- Follow the below output format.
- ...

# Output Format
<Desired output format>

# Examples
- <Example text> <classification>
- ...
```

Figure: General Prompt Structure - Part 2